

### • ABOUT ME

Full-stack developer with an MSc in Geo-Information Science, combining web development with a interest in data and geospatial technologies. On the frontend, I work with React, Next.js, and TypeScript; on the backend, Python and Node.js with REST APIs and PostgreSQL. I've applied this stack to data-intensive, real-world products from interactive web apps to internal tools enjoying the full path from data to deployment, with growing interest in applying AI/ML to practical features.

### • SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Java

**Frontend:** React, Next.js, HTML, CSS, Tailwind CSS

**Backend:** Node.js, Django REST Framework, REST APIs, Celery, Redis

**Databases:** PostgreSQL/PostGIS, MongoDB (schema design, indexing, query optimization)

**Geospatial:** GDAL, Rasterio, GeoPandas, Shapely, OpenLayers, MapLibre GL

**AI / ML:** PyTorch, TensorFlow, scikit-learn, OpenCV, OpenAI API

**DevOps & Tools:** Docker, Git/GitHub, Linux, AWS, CI/CD

### • EXPERIENCE

#### ML Assistant

June 2026 – July 2026

**Earth and Life Institute, Université catholique de Louvain** – Louvain-la-Neuve, Belgium

- Applied field boundary delineation AI model developed in my thesis to generate 2026 agricultural field boundary maps for the Wallonia region, Belgium
- Processed and fused Satellite image time series data for the Walloon agricultural landscape to support regional-scale delineation

#### Geospatial AI Internship

May 2025 – Aug 2025

**Estonian Land and Spatial Development Board** – Tartu, Estonia

- Built end-to-end raster preprocessing data in Python to ingest and normalise national-scale aerial imagery, handling large datasets spanning the entire country.
- Developed a semantic segmentation inference pipeline for rooftop solar panel detection, transforming aerial imagery into model-ready inputs and converting outputs into structured data deliverables.
- Researched machine learning approaches to estimate building height from oblique aerial imagery, using window height as a geometric cue.

#### FullStack GIS Developer

May 2023 – Aug 2024

**Seed Innovation Private Limited** – Kathmandu, Nepal

- Designed and maintained PostgreSQL data models for agriculture insurance datasets, focusing on data quality, schema design, and performance through indexing and query optimization.
- Built Python (Django REST Framework) and REST API workflows for data validation, filtering, and geometry checks.
- Built interactive dashboards and web maps (React, OpenLayers, MapLibre GL) for internal and client-facing use.
- Processed satellite and field data (vegetation indices, aggregation) to produce operational data products for planning workflows.
- Offloaded vegetation-index calculations and bulk data analysis to background workers (Celery + Redis), keeping the web app responsive during large uploads.

- Developed interactive web interfaces for data visualization and analysis.
- Developed reusable react components for map applications using MapLibreGL and OpenLayers to publish interactive, shareable geospatial visualizations
- Collaborated with developers and domain specialists to turn requirements into working application features.

## • EDUCATION

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02/09/2024 – 26/6/2026

### **GEO-INFORMATION SCIENCE (MSc)**

Erasmus Mundus Joint Master (Double degree)

University of Tartu, Tartu, Estonia (year 1 2024 - 2025)

Université catholique de Louvain, Louvain-la-Neuve, Belgium (year 2 2025 -2026)

08/2018 – 05/06/2023 Dhulikhel, Nepal

### **GEOMATICS ENGINEERING (B.SC)** Kathmandu University

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## • ONLINE PROFILES

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LinkedIn: [linkedin.com/in/sushant-aryal-56360b1a4](https://www.linkedin.com/in/sushant-aryal-56360b1a4)

GitHub: [github.com/SushantAryal3](https://github.com/SushantAryal3)

## • LANGUAGES

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English (Proficient C1), French (A1), Nepali (fluent)